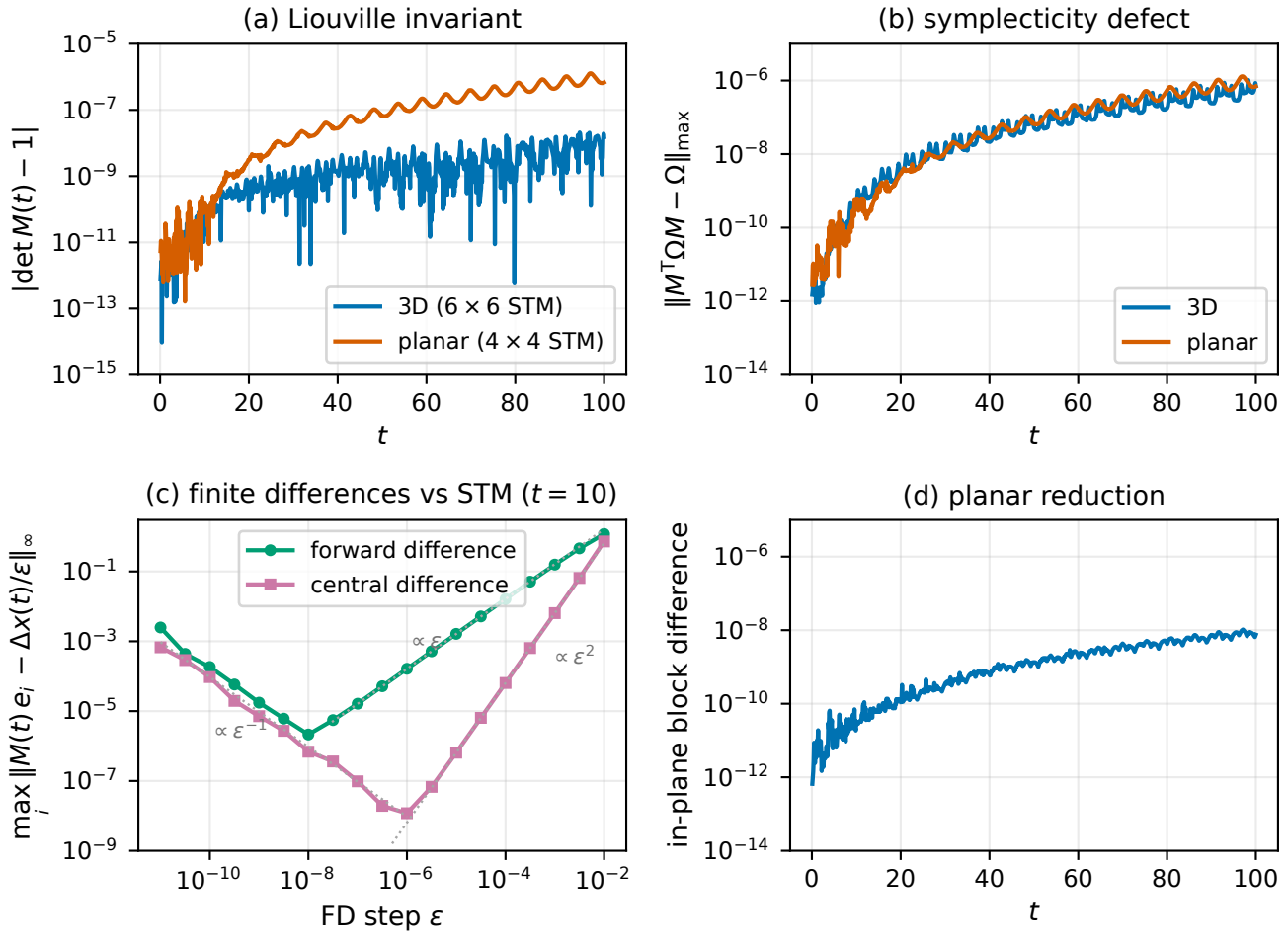
(a) Liouville invariant
$|\det M(t) - 1|$
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
3D (6 × 6 STM)
planar (4 × 4 STM)
0
20
40
60
80
100
t
(b) symplecticity defect
$\|M^{\mathsf{T}}\Omega M - \Omega\|_{\max}$
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
3D
planar
0
20
40
60
80
100
t
(c) finite differences vs STM ($t = 10$)
$\max_i \|M(t)\, e_i - \Delta x(t)/\varepsilon\|_\infty$
forward difference
central difference
$\propto \varepsilon$
$\propto \varepsilon^{-1}$
$\propto \varepsilon^{2}$
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-10}
10^{-8}
10^{-6}
10^{-4}
10^{-2}
FD step ε
(d) planar reduction
in-plane block difference
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
0
20
40
60
80
100
t